PID-RS MATHEMATICAL PUBLICATION MAP

Mathematical results in pid-rs

Theorems, evidence, costs, boundaries, and open obligations

A scoped guide to eight result families. Each result card states its mathematical object, assumptions, central formula, evidence, cost, use, and strongest nonclaim.

The Markdown source is canonical. Scope-specific reports, claim packets, and the method catalog remain authoritative for their stated domains.

28 August 2026

Canonical source: `MATHEMATICAL_RESULTS_GUIDE.md`

Contents

1. Reading conventions and semantic firewall **2**
Evidence labels . . . 2
Five distinct lanes . . . 2
Lattice positions versus audit coordinates . . . 3

2. Result map **5**

3. Categorical-Sx theory **7**
3.1 Foundational semantic audit . . . 7
3.2 Fixed finite-alphabet plug-in convergence . . . 7
3.3 Support-change-tolerant averaged-Sx continuity . . . 9

4. Sampling and exact finite-table assurance **10**
4.1 Dependency-color concentration . . . 10
4.2 Exact two-source categorical-Sx assurance . . . 11

5. Higher-source, numerical, and continuous-estimator assurance **12**
5.1 SxPID3 source-marginal factorization and bounded audit . . . 12
5.2 Represented-binary64 and quantizer assurance . . . 13
5.3 KSG positive-integer harmonic arithmetic . . . 14

6. Estimator choice, global nonclaims, and further reading **15**

This guide gives junior readers a map of the principal mathematical results and assurance work in pid-rs. Each result card identifies the mathematical object and its assumptions. Each card also gives the central formula, evidence, cost, use, strongest nonclaim, and complete proof or publication artifact.

Non-authoritative guide. Scope-specific files govern each type of statement:

- `method-catalog.json` governs method identity, provenance, implementation status, and the public surface.
- `METHODS.md` and `METHODS_SUMMARY.md` provide generated views of the catalog.
- The linked active revision index and decision govern the claim lifecycle.
- The detailed Markdown and PDF reports govern exact statements.

If this guide conflicts with a scope-specific governing source, the governing source takes precedence.

The repository assigns authority by scope. The narrow SxPID3 factorization/bounded audit is an integrated stable validation entry. The separate full SxPID3 certificate specification remains proposed and unsupported. Neither status changes the other.

In this guide, "New in pid-rs" means repository implementation, rederivation, theorem composition, counterexample, diagnostic, or assurance work. The phrase is **not** a scientific-priority claim. All information quantities below use natural logarithms and are in **nats**. The MGW paper uses bits. Multiplication by the positive factor $\log 2$ converts its values to nats without changing signs or equalities.

1. Reading conventions and semantic firewall

Evidence labels

Label	Meaning	Boundary
[P]	Published definition or theorem	Does not prove the local transcription or code
[R]	Exact repository proof or rederivation	No scientific-priority, runtime, or statistical claim
[X]	Exact counterexample	Refutes only the stated stronger claim
[B]	Bounded exhaustive/finite-corpus result	No conclusion outside its declared domain
[E]	Executable observation/test	Not a proof of its specification or runtime
[O]	Open obligation or prohibited transfer	Does not negate a narrower accepted result

A **functional** takes a probability law as its input. An **estimator** infers a functional from samples. A PID **cumulative** belongs to a lattice node. Möbius inversion produces an **atom** from the cumulatives.

Exact real, exact rational, and represented binary64 statements are different types of statements. Formal and executable routes can share human statements or transcriptions. Therefore, route counts do not imply independence.

Five distinct lanes

Lane	Object in this repository	Do not conflate it with
Categorical MGW shared exclusions	Finite-law event-logical PID of Makkeh, Gutknecht, and Wibral (2021), implemented for two to four sources	Ehrlich continuous Sx, Williams–Beer $I_{\min}$, or BROJA
Continuous Ehrlich shared exclusions	Distinct continuous functional/kNN estimator of Ehrlich et al. (2024). Experimental here	A categorical theorem or transparent quantization of the categorical theorem
Williams–Beer $I_{\min}$	Different categorical redundancy of Williams and Beer (2010)	Categorical Sx merely because both use an antichain carrier
BROJA	Bivariate finite-alphabet optimization-based PID proposal of Bertschinger et al. (2014). Comparison boundary only here	Any accepted mapping from Sx or $I_{\min}$. The repository asserts no mapping
KSG	MI estimator of Kraskov, Stögbauer, and Grassberger (2004)	A PID definition or proof of a downstream continuous PID

Figure 1: Semantic firewall for the five non-interchangeable method lanes.

This guide therefore avoids the blanket phrase "Wibral PID." The repository has no accepted result for either of these ideas:

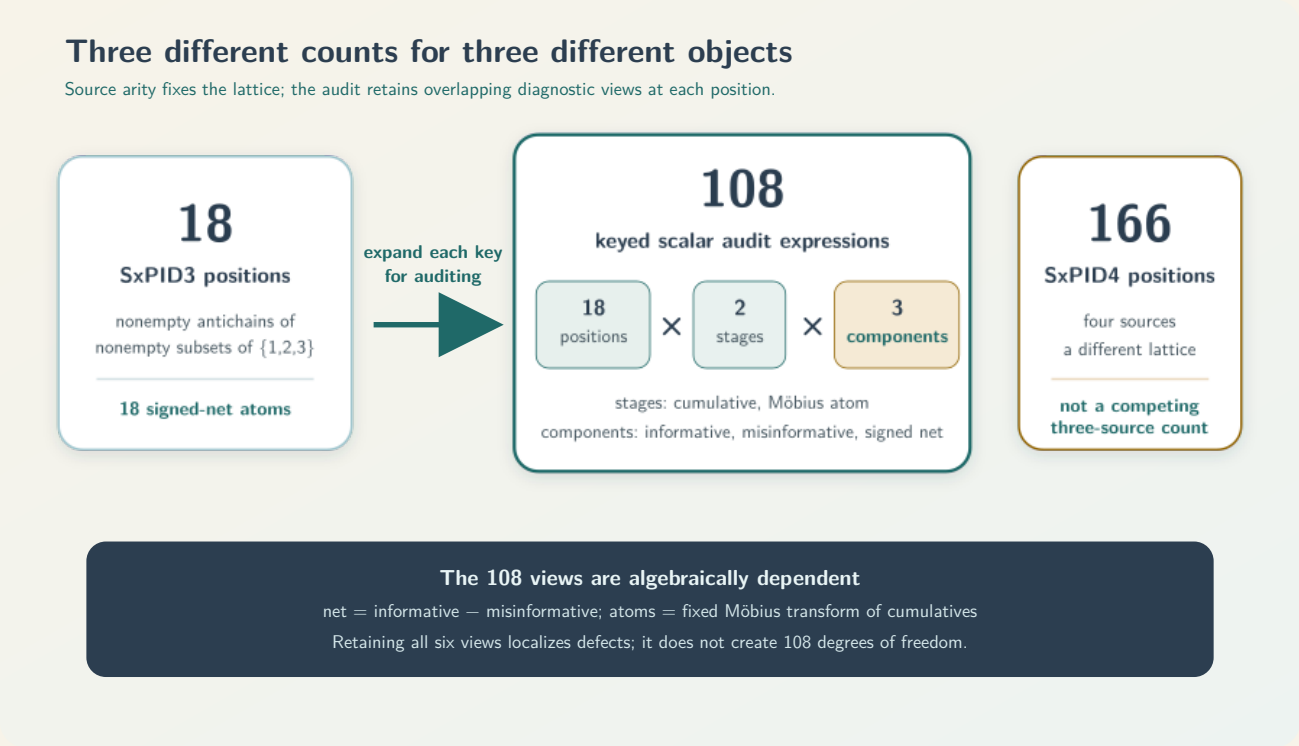
- A target-permutation affine-reflection theorem.
- A generic affine reconstruction across different PID definitions.

Lattice positions versus audit coordinates

Categorical Sx has 4, 18, and 166 lattice positions for two, three, and four sources. Audit registries expand each lattice by two stages (cumulative/atom) and three components (informative/misinformative/net):

$$24 = 4 \times 2 \times 3, \qquad 108 = 18 \times 2 \times 3.$$

Thus, SxPID3 has 18 net atoms. The count 108 is an audit-registry coordinate count. The count 166 belongs to SxPID4.


Three different counts for three different objects
Source arity fixes the lattice; the audit retains overlapping diagnostic views at each position.
18
SxPID3 positions
nonempty antichains of nonempty subsets of {1,2,3}
18 signed-net atoms
expand each key for auditing
108
keyed scalar audit expressions
18 positions
×
2 stages
×
3 components
stages: cumulative, Möbius atom
components: informative, misinformative, signed net
166
SxPID4 positions
four sources
a different lattice
not a competing three-source count
The 108 views are algebraically dependent
net = informative − misinformative; atoms = fixed Möbius transform of cumulatives
Retaining all six views localizes defects; it does not create 108 degrees of freedom.

2. Result map

Family	Status and evidence	Detailed source
1. Foundational categorical-Sx audit	Stable validation. Published object [**P**]. Repository audit [**R,X,B,E**]. Broad semantics open [**O**]	`FOUNDATIONAL_SHARED_EXCLUSIONS_PID_AUDIT.md` · PDF
2. Finite plug-in convergence	Stable validation. Published limit tools [**P**]. Repository composition [**R,X,B,E**]. Floating/runtime bridge open [**O**]	`FINITE_ALPHABET_PLUGIN_CONVERGENCE.md` · PDF
3. Support-change averaged-Sx continuity	Stable validation. Theorem/counterexamples [**R,X**]. Bounded/formal subevidence [**B,E**]. Full refinement open [**O**]	`SUPPORT_CHANGE_TOLERANT_AVERAGED_SXPID_CONTINUITY.md` · PDF
4. Dependency-color concentration	Stable validation. Published inequalities [**P**]. Repository composition/Sx bounds [**R,X,B,E**]. Automatic coloring absent [**O**]	`DEPENDENCY_COLORED_SXPID_CONCENTRATION.md` · PDF
5. Exact SxPID2 assurance	Count-to-atom formal scope complete [**R**]. Product-one counterexample [**X**]. Certifier conditional [**B,E**] with integration open [**O**]	count-to-atom proof · exact-product report · certifier decision v3
6. SxPID3 factorization/audit	Integrated stable validation: method anchor [**P**], factorization [**R**], counterexamples [**X**], bounded audit [**B,E**], correspondence/certificate open [**O**]	`SXPID3_SOURCE_MARGINAL_AND_BOUNDED_AUDIT.md` · PDF
7. Binary64/quantizer assurance	Implemented measure-specific arithmetic/diagnostics [**R,B,E**]. Bounded nonfindings and policy rejections [**B,E,O**]. No estimator or full refinement [**O**]	`NUMERICAL_ASSURANCE.md`. Dedicated PDF absent
8. KSG integer-harmonic arithmetic	Exact/formal/bounded core scoped GO [**P,R,X,B,E**]. Repository/publication integration **NO-GO** [**O**]	`claim-v4.md` · `integration-disposition-v4.md`

MATHEMATICAL RESULT LEDGER

Eight families, four evidence questions

Completion is stated per row; proximity is not proof transfer.

analytic theorem / exact analysis · kernel- or solver-checked subclaim · bounded corpus / replay / fixture · document or integration state

RESULT FAMILY exact scope named in every row	ANALYTIC what is proved on paper?	FORMAL which subclaims are checked?	BOUNDED EXECUTABLE what finite evidence ran?	REPORT / INTEGRATION not a mathematical proof level
1 **Foundational categorical Sx audit**	Exact semantics + counterexamples	Lean semantic factorization slice	Exact witnesses + bounded checks	SCOPED DETAILED REPORT Markdown + PDF
2 **Finite-alphabet plug-in convergence**	Exact-real convergence theorem	Lean deterministic continuity core	Finite fixtures + Rust comparison	SCOPED DETAILED REPORT Markdown + PDF
3 **Support-change-tolerant averaged Sx continuity**	Exact-real continuity moduli	Lean algebraic subclaims	18 law pairs + bounded replay	SCOPED DETAILED REPORT Markdown + PDF
4 **Dependency-color concentration and local bounds**	Probability theorem + local Sx bounds	Lean deterministic subclaims only	Oracle + Rust bounded checks	SCOPED DETAILED REPORT Markdown + PDF
5 **Exact two-source Sx assurance**	Count → atoms + product theorem	Lean bridge + product algebra	Exact certifier + bounded routes	THREE DISTINCT LAYERS Multiple scoped PDFs Certifier conditional Integration open
6 **SxPID3 source-marginal factorization and bounded audit**	Source-marginal factorization	Lean generic finite-event theorem	20,348 tables two exact routes	INTEGRATED • SCOPED Markdown + PDF Full certificate remains open
7 **Represented-binary64 and quantizer assurance**	Exact represented operand reduction	No end-to-end formal proof claimed	Exact oracles + bounded tests	CANONICAL MARKDOWN Dedicated PDF open
8 **KSG integer-harmonic local arithmetic**	Harmonic identity + sharp box bound	Lean + Z3 scoped arithmetic	8,198-row bounded corpus	SCOPED CORE: GO Exact • formal • bounded Integration: NO-GO Dedicated result PDF open

READ THE CELLS LITERALLY

Formal subclaims, exact arithmetic, bounded corpora, report PDFs, and integration gates remain separate evidence objects.

Figure 2: Evidence and publication status map for the eight result families.

`stable` is a catalog family status. The status does not claim estimator consistency, calibration, application validity, scientific novelty, or complete formal verification.

3. Categorical-Sx theory

3.1 Foundational semantic audit

Scope and formula. The formula uses a supported finite key $z = (s, t)$ and an antichain α. The symbol p denotes the finite probability law. The symbol $A_\alpha(s)$ denotes the MGW disjunction of matching source events. The cumulative components are:

$$c_\alpha^+ = -\log p(A_\alpha), \qquad c_\alpha^- = \log \frac{p(T = t)}{p(T = t, A_\alpha)}, \qquad c_\alpha^{\text{net}} = c_\alpha^+ - c_\alpha^-.$$

The symbols c_α^+, c_α^-, and c_α^{net} denote informative, misinformative, and net cumulative components, respectively.

The full-lattice atoms are the fixed finite Möbius inverse of these cumulatives. Rota (1964) supplies the classical tool. pid-rs did not invent Möbius inversion.

Result and boundary. The audit separates native Sx properties from imported desiderata. The audit retains exact failures of these properties:

- Generic identity/local positivity.
- Source-target symmetry.
- Coarse-graining monotonicity.
- Pairwise identifiability.
- Descriptor substitution.

Published informative and misinformative component atoms are nonnegative on the stated full lattice. The net difference may be negative.

Exact witnesses and a generic Lean descriptor-factorization lemma do not prove all laws, Rust refinement, causality, or a unique interpretation. **[P,R,X,B,E,O]**

Cost/use. The result adds no new estimator. Sampled use requires the empirical full joint PMF. Lattice/event cost grows sharply from 4 to 18 to 166 nodes.

In neural or sensor work, a negative signed atom does not by itself prove harm, suppression, or common deterministic information. A separate domain theorem must support any such label.

Read next. `FOUNDATIONAL_SHARED_EXCLUSIONS_PID_AUDIT.md` and the corresponding PDF. Catalog entry: `validation.foundational-shared-exclusions-audit`.

3.2 Fixed finite-alphabet plug-in convergence

Assumptions/result. The result fixes a finite alphabet, a lattice, and a law P with support S. The sequence of laws Q_n must satisfy both conditions:

- $Q_n(z) \to P(z)$ for every cell.
- $\text{supp}(Q_n) \subseteq S$.

The supports of Q_n eventually equal S. All supported-key categorical-Sx cumulatives/atoms/averages then converge. Separate continuity arguments cover the listed $I_{\min}$ and Shannon quantities without equating their definitions.

The premises hold almost surely in either of two cases:

- The rows are i.i.d. with marginal P.
- The rows are strictly stationary **ergodic** with marginal P.

Stationarity alone is insufficient. **[P,R,X]**

For i.i.d. rows on K cells, α is a failure-probability budget with $0 < \alpha < 1$. A Hoeffding/spending-sequence composition gives the following simultaneous bound for every prefix n with probability at least $1 - \alpha$:

$$\|\hat{P}_n - P\|_1 \leq B_n := K\sqrt{\frac{\log(K\pi^2 n^2/(3\alpha))}{2n}}.$$

Here, $\hat{P}_n$ is the empirical law of the first n rows. The quantity B_n is the simultaneous L^1 radius.

A guaranteed support-entry time also needs a known population floor $p_{\min} = \min_{z \in S} P(z) > 0$. An empirical minimum cannot reveal an unseen rare cell.

The frozen-transform corollary uses the training sigma-field $\mathcal{G}$. The training artifact and fitted transform generate $\mathcal{G}$. The corollary requires all these conditions:

- The raw evaluation space is standard Borel.
- The failure symbol $\perp$ is outside the finite output alphabet $\mathcal{Z}$.
- One map $Q(\omega, x)$ is measurable from $\mathcal{G} \otimes \mathcal{B}(\mathcal{X})$ to $\mathcal{Z} \cup \{\perp\}$.
- The training sigma-field determines the random choice of Q.
- The output alphabet and block structure are finite and fixed.
- The map is frozen for every evaluation row and every prefix.
- Evaluation rows are conditionally i.i.d. given $\mathcal{G}$.
- $\Pr(Q(W_1) \neq \perp \mid \mathcal{G}) = 1$ almost surely.

Here, $\mathcal{B}(\mathcal{X})$ is the Borel sigma-field on the raw evaluation space. The symbol W_1 denotes one raw evaluation row.

Conditional on $\mathcal{G}$, the target is the random push-forward law $P_Q(z) = \Pr(Q(W_1) = z \mid \mathcal{G})$. It is generally not the unconditional mixture law. The theorem applies to the conditional functional for almost every training realization.

An i.i.d. raw evaluation sequence independent of the training artifact is an important special case. Otherwise, the corollary does not require independence from the training artifact after the conditional-i.i.d. contract is proved.

This theorem-level statement is broader than the stable fitted-wrapper contract. The stable fitted-quantized APIs require the training artifact to be independent of the raw evaluation sequence. This guide does not relax that cataloged API condition.

Evidence/cost/use. Lean covers a deterministic core. Lean does not cover the stochastic theorem, complete higher-source semantics, Rust, or floating point. The bound can be conservative.

This result gives a consistency route for an empirical plug-in estimator. The result is not a new estimator or calibration theorem. The corollary covers a held-out quantizer application only when all frozen-transform conditions hold. Quantization changes the estimand. **[R,B,E,O]**

Read next. `FINITE_ALPHABET_PLUGIN_CONVERGENCE.md` and the corresponding PDF. Catalog entry: `validation.finite-alphabet-plugin-convergence`.

3.3 Support-change-tolerant averaged-Sx continuity

Assumptions/result. The result fixes three objects:

- One complete finite alphabet of size K.
- The source-event family.
- The full Möbius lattice.

Laws p, q may create/delete support inside that alphabet. The definitions are $\eta = \frac{1}{2}\|p - q\|_1$, $r_z = \min(p_z, q_z)$, $a = p - r$, and $b = q - r$. The result also uses $E_\vee = \max(E(a), E(b))$ and $E_\Sigma = E(a) + E(b)$, where $E(d) = -\sum_{d_z > 0} d_z \log d_z$.

The quantity η is the total variation distance. The vector r contains the shared cellwise mass. The vectors a and b contain the two residual masses. The function E is the displayed residual-mass sum.

The function is $g_J(\eta) = (1-\eta)\log(1+J\eta/(1-\eta))$ on $0 < \eta < 1$. Its endpoint values are $g_J(0) = g_J(1) = 0$. The other definitions are $W_\alpha = \sum_\beta |M_{\alpha\beta}| g_{J_\beta}$ and $s_\alpha = \sum_\beta M_{\alpha\beta}$.

Here, $M_{\alpha\beta}$ is a coefficient of the fixed Möbius inverse. The branch count is $J_\beta = |\beta|$. The quantity s_α is a row sum of that inverse. The atom bounds are:

$$|\Delta\Pi_\alpha^+| \le E_\vee + W_\alpha, \quad |\Delta\Pi_\alpha^-| \le E_\vee + W_\alpha + |s_\alpha| g_1, \quad |\Delta\Pi_\alpha^{\text{net}}| \le E_\Sigma + 2W_\alpha + |s_\alpha| g_1.$$

For $\eta > 0$ (which implies $K \ge 2$), $E_\vee \le \eta \log((K-1)/\eta)$ and $E_\Sigma \le \eta \log(\lfloor K^2/4 \rfloor / \eta^2)$. Both envelopes take the value zero at $\eta = 0$.

The exact functions above are not all monotone on $[0, 1]$. Suppose a statistical result gives only $\eta \le \varepsilon$. Direct substitution of ε into the exact formulas is invalid without a monotonicity proof.

For $K \ge 2$, the monotone upper envelopes are:

$$\bar{e}_K^\vee(\varepsilon) = \varepsilon \left[1 + \log \frac{K-1}{\varepsilon}\right], \qquad \bar{e}_K^\Sigma(\varepsilon) = \varepsilon \left[2 + \log \frac{\lfloor K^2/4 \rfloor}{\varepsilon^2}\right].$$

Both barred envelopes take the value zero at $\varepsilon = 0$. For $0 \le \eta \le \varepsilon \le 1$, the valid replacements satisfy:

$$E_\vee \le \bar{e}_K^\vee(\varepsilon), \qquad E_\Sigma \le \bar{e}_K^\Sigma(\varepsilon), \qquad g_J(\eta) \le J\eta \le J\varepsilon.$$

These monotone replacements can feed the atom bounds.

Evidence/boundary. The repository proves the residual/load decomposition. Fixed-system witnesses establish a lower limit for one specified family of bounds. This family uses one common leading coefficient and covers the retained witnesses. Such a family needs a coefficient of at least 1 for components and 2 for net atoms.

The result does not include any of these theorems:

- A global linear theorem.
- A pointwise disappearing-key theorem.
- An alphabet-free theorem.
- An active-face Fannes theorem.
- A truncated-lattice theorem.

Lean covers subclaims. Lean does not cover the complete probability/Rust bridge. **[R,X,B,E,O]**

Cost/use. The result adds neither an estimator nor a confidence interval. A separately justified law-distance radius can feed the deterministic modulus through the barred replacements above. Fixed lattice coefficients are precomputable.

The result applies when rare cells enter/leave while the alphabet and quantizer remain fixed.

Read next. `SUPPORT_CHANGE_TOLERANT_AVERAGED_SXPID_CONTINUITY.md` and the corresponding PDF. Catalog entry: `validation.support-change-tolerant-averaged-sxpid-continuity`.

4. Sampling and exact finite-table assurance

4.1 Dependency-color concentration

Assumptions/result. Fix an integer $n \geq 1$. Complete rows $Z_i = (S_{1i}, \dots, S_{mi}, T_i)$ for $1 \leq i \leq n$ share one law P on a K-cell alphabet with $K \geq 2$. The design must declare a coloring κ before outcomes are observed. The set of color labels is finite or countable. Rows must be jointly mutually independent **within** each color.

Dependence across colors may be arbitrary. Dependence among coordinates within each complete row is unrestricted.

None of these weaker conditions is sufficient:

- Pairwise independence.
- Zero covariance.
- Adaptive colors.
- An unspecified mixing label.

The class load $n_{a,n}$ counts the first n rows that have color a. The following bound holds for every $\varepsilon > 0$. The sums run over occupied colors:

$$V_n = \left(\sum_a \sqrt{n_{a,n}} \right)^2, \qquad \Pr(\|\hat{P}_n - P\|_1 \geq \varepsilon) \leq \min \left\{ 1, (2^K - 2) e^{-n^2 \varepsilon^2 / (2V_n)} \right\}.$$

V_n is Janson's cover-specific factor. The factor is not an estimated effective sample size. The derivation uses Hoeffding, Janson, and Weissman et al. Repository work supplies prefix/drift and Sx-specific compositions.

A separate common-support local theorem uses laws p and q on the same fixed complete finite alphabet. It has both law conditions:

- $\operatorname{supp}(q) \subseteq \operatorname{supp}(p)$.
- $\delta = \|q - p\|_1 < 2p_{\min}$.

Here, $p_{\min} = \min_{z \in \operatorname{supp}(p)} p(z) > 0$.

The local theorem uses a supported realization $z = (s_1, \dots, s_m, t)$. It also uses a nonempty antichain α of nonempty source subsets. For each component $u \in \{+, -, \mathrm{sx}\}$, the theorem states:

$$|c_\alpha^u(z; q) - c_\alpha^u(z; p)| \leq \Lambda, \qquad \Lambda = \log \frac{p_{\min}}{p_{\min} - \delta/2}.$$

The label sx denotes the net cumulative component. Thus, Λ bounds the absolute between-law change, not the cumulative value. **[P,R,X]**

Evidence/cost/use. Lean checks deterministic algebra. Lean does not check whether a scientific design satisfies independence. Computing V_n is cheap. However, $2^K - 2$ can make the bound vacuous, and validation of the dependence premise can be difficult. pid-rs supplies no public automatic color estimator.

A fixed-width transformation of i.i.d. innovations can satisfy the premise. The coloring must be predeclared, and same-color windows must use disjoint innovations. Arbitrary rolling data does not satisfy the premise merely because a transformation has fixed width. **[B,E,O]**

Read next. `DEPENDENCY_COLORED_SXPID_CONCENTRATION.md` and the corresponding PDF. Catalog entry: `validation.dependency-color-sxpid-concentration`.

4.2 Exact two-source categorical-Sx assurance

Assumptions/result. Exactly two finite categorical sources and a finite target have natural counts c_z with $n = \sum_z c_z > 0$. The conditions allow zero cells. The conditions prohibit smoothing. For every cumulative or atom coordinate, exact event counts give a positive rational R with

$$V = \frac{1}{n} \log R, \qquad V = 0 \iff R = 1, \qquad \operatorname{sign}(V) = \operatorname{sign}(R - 1).$$

The assurance covers the 24 coordinates $4 \times 2 \times 3$. A retained counterexample shows that a nonempty term map may still have product one. **[R,X]**

Exact-product preflight. A canonical expression has positive rational bases x_j and rational coefficients q_j with $nq_j \in \mathbb{Z}$. Its denominator-cleared product and conservative projection are:

$$R = \prod_j x_j^{nq_j}, \qquad B = \sum_j |nq_j| \left(\operatorname{bits}(\operatorname{num} x_j) + \operatorname{bits}(\operatorname{den} x_j)\right).$$

Here, num and den give the numerator and denominator. The function bits gives the integer bit length.

Status 1: count-to-atom bridge. The Lean bridge is complete only in its supplied-count, fixed-two-source scope. It does not prove paper correspondence, component nonnegativity, decoding, Rust/binary64, sampling, or higher sources.

Status 2: exact-product path. One expression has at most 256 terms. Each absolute cleared exponent is at most 16,384. Per-expression projected bits satisfy $B \leq 262{,}144$. The aggregate projection is at most 1,048,576. A rejection means `unavailable` and never supplies a sign.

Status 3: conditional certifier. Certifier revision 3 gives conditional assurance for accepted interval reports. It gives an exact sign or zero only when status is `compared`. Integration and independent custody remain open. The intervals are not confidence intervals. **[R,B,E,O]**

Cost/use. Big rational products can grow quickly. Therefore, the exact-product path is an offline oracle/escalation path for small empirical tables or near cancellation. The exact-product path is not a population estimator.

Read next: count-to-atom proof. See `TWO_SOURCE_SXPID_COUNT_ATOM_BRIDGE.md` and its PDF.

Read next: exact-product report. See `EXACT_LOG_PRODUCT_SXPID2_ASSURANCE.md` and its PDF.

Read next: lifecycle decisions. See the `count-bridge decision v2` and `certifier decision v3`.

5. Higher-source, numerical, and continuous-estimator assurance

5.1 SxPID3 source-marginal factorization and bounded audit

Factorization. The factorization assumes:

- Finite source/target alphabets.
- One supplied source-only shared-exclusions event family **intended to transcribe MGW**.
- Natural-log units.
- One literally fixed transform.

"Source marginal" means the joint law P_S of (S_1, S_2, S_3). The term does not mean three separate marginals. The set $E_\alpha(s)$ is the supplied source-only shared-exclusions event for key s and node α. For supported source keys, the informative cumulative is:

$$I_\alpha^+(P) = \sum_{s:P_S(s)>0} P_S(s) \left[-\log \sum_{s' \in E_\alpha(s)} P_S(s') \right] = G_\alpha(P_S).$$

Hence, equal complete source marginals give equal informative cumulatives. Applying one fixed linear transform to equal cumulatives gives equal transformed coordinates. Equal complete source marginals need **not** preserve misinformative/net components. Exact counterexamples also show that equal separate one-source marginals do not determine the informative vector. The continuity specialization is deterministic, not a permutation calibration. **[P,R,X]**

Bounded audit. The exact domain has:

- Three ordered binary sources.
- One binary target.
- Every labeled 16-cell count table with $1 \le N \le 5$.

$$\sum_{N=1}^{5} \binom{N+15}{15} = 20{,}348, \qquad 20{,}348 \times 108 = 2{,}197{,}584.$$

The audit evaluates 2,197,584 products per route. Under a source-bound execution receipt, two routes emit the same route-neutral v2 expression-stream SHA-256. The routes also emit the same six-block sign/zero census. The routes use disjoint implementations under shared semantics. Therefore, the routes are not independent proofs. Both routes retain the human transcription and host/runtime premises. **[B,E,O]**

Status/cost/use. The catalog marks the narrow entry `validation.sxpid3-source-marginal-bounded-audit` as integrated and stable. Paper-to-local correspondence remains open. These additional obligations also remain open:

- Concrete carrier/order proofs.
- Nonzero-log enclosure.
- Compiled-Rust/binary64 refinement.
- Larger domains.
- Population validity.
- External review.

The result adds no estimator. Cached source-event masses let target-only reallocations reuse the informative block. A surrogate test still needs exchangeability. The exact audit is expensive. Big-integer size depends on the input.

Read next. `SXPID3_SOURCE_MARGINAL_AND_BOUNDED_AUDIT.md` and PDF. The separate full-certificate `revision index` and `proposed decision` have no accepted target evidence. The separate full-certificate status does not downgrade or strengthen the integrated narrow result.

5.2 Represented-binary64 and quantizer assurance

Result. Every finite binary64 number is an integer multiple of 2^{-1074}. On supported 32-bit and 64-bit targets, each positive or negative magnitude array has $L = 34$ fixed limbs. Thus, one accumulator has 68 limb slots in its two magnitude arrays. For finite operands, the reducer returns:

$$\mathrm{reduce}(x_1, \dots, x_n) = \mathrm{RN}_{\mathrm{even}}\left(\sum_i x_i\right).$$

The operator $\mathrm{RN}_{\mathrm{even}}$ rounds to nearest in binary64 and breaks ties to even. The inner sum is exact for the **already represented** operands. The measure-specific call sites are:

- Final averaging of represented categorical-Sx informative and misinformative components. The net component remains their derived difference.
- The four-operand Williams–Beer $I_{\min}$ PID2 synergy.
- Represented continuous-PID2 synergy plus its three reconstruction checks.

The reducer does not make probabilities, logarithms, or estimates exact. The reducer transfers no semantics between those PID families. The reducer rejects a non-finite input. The reducer also rejects an accepted-operand count beyond `usize::MAX`. Exact cancellation returns positive zero. Estimator-facing callers reject an exact sum that rounds beyond the finite binary64 range. **[R,B,E]**

The equal-width quantizer now constructs finite monotone edges across extreme ranges. The quantizer reports nominal and finite-map-reachable joint cardinalities as `Option<u128>` values. The quantizer reports the observed joint cardinality as `usize`. An optional numeric value is present only when the relevant product fits in `u128`. `None` denotes overflow of that optional product. Map reachability is not population support.

Bounded investigations did not justify these changes:

- Changing pointwise Sx Möbius reduction.
- Changing three-source $I_{\min}$ reduction.

These are bounded executable nonfindings, not counterexamples. **[B,E,O]**

The repository also rejected an order-dependent fast mode as policy-incompatible. **[O]**

Cost/use. For n operands, reduction is $O(nL)$ with fixed $L = 34$. The resource envelope charges at most $2L$ limb visits per accepted add and $4L$ visits for finalization. Two Sx component accumulators per lattice node require about:

- 4,416 bytes for two sources.
- 19,872 bytes for three sources.
- 183,264 bytes for four sources.

These amounts exclude other structures. Suitable uses include reproducible channel ordering, exact reachability diagnostics, and resource preflight. The assurance is not a portable latency or estimator-error theorem.

Read next. `NUMERICAL_ASSURANCE.md`. A dedicated `output/pdf/numerical-assurance.pdf` does not exist. This guide does not replace a dedicated Numerical Assurance PDF.

5.3 KSG positive-integer harmonic arithmetic

Status and theorem. Revision 4 is active. The exact/formal/bounded core has scoped GO results. However, repository/publication integration remains **NO-GO**. Final `decision-v4.md` and `evidence-matrix-v4.md` are absent.

The symbol ψ denotes the digamma function, and γ denotes the Euler–Mascheroni constant. For every positive integer m, NIST DLMF Equation 5.4.14 gives $\psi(m) = H_{m-1} - \gamma$. Here, $H_0 = 0$ and $H_j = \sum_{r=1}^{j} 1/r$ for $j \geq 1$. The variables x and y are positive-integer digamma arguments from the inventoried estimator mappings.

The theorem has these conditions:

- $n \geq 2$.
- $1 \leq k < n$.
- $k \leq x, y \leq n$.
- Coefficient pattern $(+1, +1, -1, -1)$.
- The published positive-integer identity is supplied as a typed premise at all four arguments.

Under these conditions, the arithmetic term satisfies:

$$T = \psi(k) + \psi(n) - \psi(x) - \psi(y) = H_{k-1} + H_{n-1} - H_{x-1} - H_{y-1}, \qquad -D \leq T \leq D.$$

Here, $D = H_{n-1} - H_{k-1}$. The bound is sharp on the rectangular arithmetic domain. Neighbor geometry need not fully realize that domain. A stronger $x + y \leq n + k$ route remains an unpromoted conditional source lemma. **[P,R,O]**

Evidence/boundary. The packet has 19 scoped Lean theorems and 4 conditional Z3 obligations. The packet also has bounded 8,198-row modular/enclosure/compiled evidence. Routes share the analytic premise and human statement choices.

A rejected prime has exact-nonzero zero-residue collisions. Thus, modular zero is not exact zero in general. Selected primes provide redundant fault diversity, not independent proofs.

Nothing here proves:

- Rust refinement.
- Neighbor geometry.
- KSG consistency.
- Ehrlich validity.
- PID atoms. **[X,B,E,O]**

Cost/use. A shifted harmonic table costs $O(n)$ setup/storage and gives $O(1)$ eligible local terms. The table is a strong arithmetic regression oracle. The table is not a new estimator, speed guarantee, support theorem, or calibration result.

Read next. `revision-index.md`, `claim-v4.md`, `formal-assurance-v4.md`, and `integration-disposition-v4.md`. The existing `ksg-m1a-composite-*` PDFs are process/boundary packets, not a self-contained math paper.

6. Estimator choice, global nonclaims, and further reading

Data/question	First route	Still required
Supplied finite PMF/counts	Direct categorical functional. Exact assurance within its scope	Correct method/event encoding
I.i.d. categorical sample	Empirical plug-in plus finite-alphabet result	Fixed law, support/sample-size and UQ arguments
Explicit dependence coloring	Dependency-color bound	Predeclared valid within-color mutual independence
Nearby finite laws with support changes	Averaged-Sx modulus	Same alphabet/event/lattice and a justified law-distance radius
Continuous sample	Report-first KSG or experimental Ehrlich surface	Support, geometry, bias/calibration, dependence, and UQ
Near-zero empirical SxPID2 value	Bounded exact-product escalation	Two-source supplied-count scope and conditional statuses

The following boundaries apply across all results:

- Bounded agreement stays bounded.
- Oracles must not read answer tables.
- Oracles must not import the implementation result as truth.
- Oracles must not weaken tolerances to pass.
- Oracles must not convert "unavailable" into a sign.
- The result boundaries prohibit clamping negative net atoms and exact zeros.
- Quantization or added noise changes the estimand.
- Shared names, citations, lattices, and software dependencies do not transfer semantics.
- No result supplies causal meaning, consumer authorization, or formal verification of all pid-rs.

Every use example above is illustrative. No use example qualifies an application.

These sources describe process and durable evidence:

- `MATHEMATICAL_PROBLEM_SOLVING_WORKFLOW.md` and the corresponding PDF.
- `PID_MATHEMATICAL_AUDIT_PROTOCOL.md`.
- `PID_DISCOVERY_VERIFICATION_AND_DURABILITY_BLUEPRINT.md`.
- `FORMAL_TOOL_ADOPTION_AUDIT.md`.
- `KNOWN_LIMITATIONS.md`.

These workflows preserve evidence. The workflows do not raise a theorem beyond its assumptions.

The guide uses these primary sources:

- The MGW categorical-Sx paper.
- The part-whole/formal-logic account.
- The Ehrlich continuous-Sx paper.
- Williams–Beer $I_{\min}$.
- Bertschinger et al. (2014) on bivariate finite-alphabet unique information.
- KSG.
- NIST DLMF Equation 5.4.14.
- Rota's Möbius theory.
- Hoeffding.
- Janson.
- Weissman et al.

The linked detailed reports give equation-level provenance. Each report marks the precise point where each source stops applying.